

REST API & Testing

Datenquelle:

<https://jsonplaceholder.typicode.com/posts>

Was brauchen wir im Node Umfeld:

- nodejs
- expressjs
- axios / fetch
- jest (test runner)
- supertest (http request testen)

Was ist das Ziel der Übung?

GET /posts

- Holt Daten von JSONPlaceholder
- Gibt sie zurück

POST /posts

- Sendet neuen Post an JSONPlaceholder
- Gibt Antwort zurück

Codebeispiel

```
import express from 'express';
import axios from 'axios';

const app = express();
app.use(express.json());

// GET /posts
app.get('/posts', async (req, res) => {
  try {
    const response = await axios.get(
      'https://jsonplaceholder.typicode.com/posts'
    );

    res.json(response.data);
  } catch (error) {
    res.status(500).json({ error: 'Fehler beim Laden der Posts' });
  }
});

// POST /posts
app.post('/posts', async (req, res) => {
  try {
    const response = await axios.post(
      'https://jsonplaceholder.typicode.com/posts',
      req.body
    );

    res.status(201).json(response.data);
  } catch (error) {
    res.status(500).json({ error: 'Fehler beim Erstellen des Posts' });
  }
});

// Server starten (nur wenn direkt ausgeführt)
if (process.argv[1] === new URL(import.meta.url).pathname) {
  app.listen(3000, () => {
    console.log('Server läuft auf http://localhost:3000');
  });
}

export default app;
```

Testing

1. Update Script in package.json

```
{
  "name": "express-api",
  "version": "1.0.0",
  "type": "module",
  "main": "app.js",
  "scripts": {
    "start": "node app.js",
    "test": "node --experimental-vm-modules node_modules/jest/bin/jest.js"
  },
  "dependencies": {
    "axios": "^1.6.0",
    "express": "^4.19.0"
  },
  "devDependencies": {
    "jest": "^29.7.0",
    "supertest": "^6.3.3"
  },
  "jest": {
    "testEnvironment": "node"
  }
}
```

2. Testdatei erstellen

/tests/posts.test.js

```
import request from 'supertest';
import app from '../app.js';

test('GET /posts', async () => {
  const res = await request(app).get('/posts');

  expect(res.statusCode).toBe(200);
  expect(res.body.length).toBeGreaterThan(0);
});
```